Curriculum Vitae

Isabella Velicogna

Position Professor

Work Address Department of Earth System Science, 3226 Croul Hall
University of California Irvine, Irvine, CA 92697-3100
Tel: -Work: 949 824 5419
Fax: 949 824 3874
e-mail: isabella@uci.edu

EDUCATION

University of Trieste, Italy	Physics	B.S., 1995
University of Trieste, Italy	Physics	M.S., 1995
University of Trieste, Italy	Engineering (Geodynamics)	Ph.D., 1999
University of Colorado, Boulder	Physics (Geodesy/Geophysics)	Postdoc, 1999-2001

EMPLOYMENT

2016-Present	Full Professor, Department of Earth System Science, University of California Irvine, Irvine, CA.
10/2018-10/2021	Associate Dean of Graduate Studies, Equity and Inclusion, School of Physical Sciences, University of California Irvine, Irvine, CA. The School of Physical Sciences has 626 graduate students, 166 tenure track faculty, 210 non-senate researchers, post-doc, specialists-, and 110 staff. It includes 4 departments: Earth System Science, Mathematics, Chemistry, Physics & Astronomy.
09/2015-09/2018	Equity Advisor, School of Physical Sciences, University of California Irvine, Irvine, CA.
2012-2016	Associate Professor, Department of Earth System Science, University of California Irvine, Irvine, CA.
2007-2012	Assistant Professor, Department of Earth System Science, University of California Irvine, Irvine, CA.
2018-Present	Visiting Scientist, Dipartimento Politecnico di Ingegneria e Architettura Università degli Studi di Udine, Udine, Italy.
2007-Present	Scientist Faculty Part Time, Division 32, NASA's Jet Propulsion Laboratory, Pasadena, CA.
2006-2007	Senior Scientist, Division 32, NASA's Jet Propulsion Laboratory, Pasadena, CA
2006-2007	Research Scientist III, Cooperative Institute for Research in Environmental Science (CIRES), University of Colorado, Boulder, CO
2004-2007	Research Scientist II, Cooperative Institute for Research in Environmental Science (CIRES), University of Colorado, Boulder, CO
2001-2004	Research Scientist I, Cooperative Institute for Research in Environmental Science (CIRES), University of Colorado, Boulder, CO

AWARDS AND RECOGNITIONS

2024	Fellow American Meteorological Society (AMS) (Fellows are selected for their outstanding contributions to the atmospheric sciences and/or their application

during a substantial period of years. Only two-tenths of one percent of AMS membership are approved as Fellows each year.).

2022 Fellow American Association for the Advancement of Science (AAAS) (Honorees have gone above and beyond in their respective disciplines. They bring a broad diversity of perspectives, innovation, curiosity, and passion that will help sustain the scientific field today and into the future. Many of these individuals have broken barriers to achieve successes in their given disciplines).

2020 Fellow of the American Geophysical Union (Fellows program was established in 1962 to recognize AGU members who have made exceptional contributions to Earth and space science through a breakthrough, discovery, or innovation in their field. A total of 62 Fellows were elected in 2020. The number of Fellows elected each year – worldwide - shall not exceed 0.1 percent of the total membership. (https://www.agu.org/Honor-and-Recognize/Honors/Union-Fellows)).

2020 Joanne Simpson Medal of the American Geophysical Union (This medal is given annually in recognition of significant contributions to Earth and space science).

2020 John Nye Lecture of the American Geophysical Union (The Nye Lecture is awarded to an outstanding cryospheric scientist in recognition of their recent accomplishments as well as their ability to present exciting science to the non-cryosphere community of AGU scientists.).

2020 Thomson Reuters Highly Cited Researchers in Geosciences (Researchers ranking among the top 1% most cited for their subject field).

2019 Thomson Reuters Highly Cited Researchers in Geosciences (Researchers ranking among the top 1% most cited for their subject field).

2018 Thomson Reuters Highly Cited Researchers in Geosciences (Researchers ranking among the top 1% most cited for their subject field).

2017 European Geosciences Union's Vening Meinesz Medal for distinguished research in Geodesy. The medal honor scientists who have achieved exceptional international standing in geodesy.

2016 Chancellor's Fellow, University of California Irvine.

2016 Thomson Reuters Highly Cited Researchers in Geosciences (Researchers ranking among the top 1% most cited for their subject field.

2015 Thomson Reuters Highly Cited Researchers in Geosciences (Researchers ranking among the top 1% most cited for their subject field).

2015 Kavli Fellow of the U.S. National Academy of Science.

2015 UCI School of Physical Sciences award for Excellence in Undergraduate Education.

2013 NASA Group Achievement Award, Ice Mass Balance Intercomparison Exercise Team (IMBIE).

2008 Kavli Fellow of the U.S. National Academy of Science.

2006 NASA Award for Outstanding Scientific Achievement.

2008 OC Metro Annual Awards for 40 under 40. This award is given to Orange County's brightest professionals who have achieved a level of success that takes some people a lifetime but are still under 40 years of age.

RESEARCH INTERESTS

Employ novel geophysical methods and satellite remote sensing techniques to understand the physical processes governing ice sheet and high mountain glacier mass balance, and the

hydrologic cycle of high latitude regions, with an emphasis on time-variable gravity from the Gravity Recovery and Climate Experiment (GRACE) missions and other geophysical data (satellite altimetry, GPS, climate model outputs, in situ observations).

Study the impact of melting ice sheets and glaciers and ice caps on global and regional patterns of sea level rise using geophysical numerical methods.

Study ecosystem response to climate change and variability in terrestrial water storage combining multiple remote sensing techniques and climate model outputs.

Study Seasonal to Sub seasonal (S2S) forecast by benchmarking climate model performances of key hydrometeorological variables based on end-user specific objectives against observations for specific region (e.g., CA), and improving regional S2S forecast personalizing a foundation AI climate model based on hyper specific local knowledge and datasets.

PUBLICATIONS

Thomson Reuters-Web of Science Researcher ID: R-5561-2018. H-index 59 (GS). 49 papers cited 100+ times; 12 cited 700+ times, 100 publications; 27,520 citations; 2,200 citations/year.
(Underlined authors are Students, Post-Doctoral Fellows and Researchers under my supervision)

PJ111- 2026 – Liang C., **Velicogna I.**, A G., Sutterley T. 'Improved GIA estimate from Gravity Altimetry and Surface Mass balance estimates.' *Submitted, GJI.*

PJ110- 2026 – Ciraci, E., **Velicogna I**., A G., Kapnick S., E. Young 'Drivers of run-off and water storage interannual variability in the Upper Indus Basin', *Submitted, Geophys. Res. Lett.*

PJ109- 2026 – Ginoux, P., S. Malyshev, E. Shevliakova, H. G. Chan, H. Guo, P.C.D. Milly, V. Naik, S. Pascale, F. Paulot, B. Pu, M. Zhao, E. Ciraci', G. A., **I Velicogna**, S.B. Kapnick, 'Distribution of absorbing aerosols in snow high mountain ranges in GFDL AM4/LM4', *Submitted, JAMES.*

PJ108- 2026 – He Z., **I. Velicogna,** Ciraci' E., Mohajerani Y., Rignot E., Mouginot, J**.**, 'Four decades of observation-based sea level fingerprint of land-ice.' *Submitted Proc. Nat. Acad. Sci.*

PJ107- 2026 – **Velicogna I.**, Sutterley T., Liang C., A G.,' Recent slowdown in ice mass loss of the Greenland and Antarctic Ice Sheets.' *Submitted Geophys. Res. Lett.*

PJ106- 2026 – Bagheri O., A G., **Velicogna I**. "Global Skillful Hotspots of S2S of S2S Weather Forecast Products in the Context of Water, Energy, and Food Nexus " *Submitted Nature Climate and Atmospheric Science.*

PJ105- 2026 – Otosaka I.N. et al., Mass balance of the Greenland and Antarctic ice sheets from the 1970s to 2023, Nature Scientific Data 13, 1301 (2026). https://doi.org/10.1038/s41597-026-08088-0.

PJ104- 2026 Pinzón, N., **Velicogna, I**., Roche, L.M. et al. 'Climate tools without climate consensus: weather and climate decision support tools create openings for agricultural adaptation', *Climatic Change*, 179, 226 (2026). https://doi.org/10.1007/s10584-026-04253-x .

PJ103- 2026 – Jha P. A G., Pathak T., **Velicogna I,** 'Accelerated climate impact on seasonal winter chill on specialty crops in California' *Nature Commun. Sustain. 1, 76 (2026). https://doi.org/10.1038/s44458-026-00084-0*

PJ102- 2026 – Shalgumbayeva G., A G., **Velicogna I**. 'Observed changes in Precipitation regime in California " *Environmental Research Communications,* 8(4), 041013.

PJ101- 2026 – Virji K. A G., **Velicogna I.** "On the importance of historical context and bias correction in deep learning MJO prediction models" *npj Clim. Atmos. Sci.* doi:10.1038/ s41612-026-01459-9

PJ100- 2025 – Rigden A., Buzzanga B, Louren M., Robinson N., **Velicogna I**., Woodborne S., 'Improved prediction of Okavango Delta inundation using GRACE-derived terrestrial water storage', *Environ. Res.: Water,* **1,** 045004, doi: 10.1088/3033-4942/ae1c67

PJ99- 2025 – Prusevich, A.A., Lammers, R., Grogan, D.S., Zuidema, S., Meko, D.D., **Velicogna, I.**, 'Decomposing land surface total water storage in the Indus, Ganges, and Brahmaputra basins'. *Frontiers in Earth Science*, 13, p.1551218.

PJ98- 2025 – Hansen J., P. Kharecha, M. Sato, G. Tselioudis, J. Kelly, S. E Bauer, R. Ruedy, E. Jeong, Q. Jin, E. Rignot, I. Velicogna, M. Schoeberl, K. von Schuckmann, J. Amponsem, J. Cao, A. Keskinen, J. Li, A. Pokela, "Global Warming Has Accelerated: Are the United Nations and the Public Well-Informed?" *Environ Sci Policy Sustain Dev 67(1):6–44. doi: 10.1080/00139157.2025.2434494*

PJ97-2024 – The GlaMBIE Team, "Community estimate of global glacier mass changes from 2000 to 2023", *Nature*, https://doi.org/10.1038/s41586-024-08545-z.

PJ96-2023 – Otosaka, I., A. Shepherd, E. Ivins, N. Schlegel, **I. Velicogna**, et al. 'Mass balance of the Greenland and Antarctic ice sheets from 1992 to 2020', *Earth Sys. Sci. Data*, **15**(4), 1597-1616.

PJ95- 2022 – P. Kishore, G. Basha, M.V. Ratnam, A. AghaKouchak, Qi. Sun, **I. Velicogna**, TBJM Ouarda, 'Anthropogenic influence on the changing risk of heat waves over India', *Nature Scientific reports* 12 (1), 3337.

PJ94- 2022 – P. Kishore, G Basha, M.V. Ratnam, A. AghaKouchak, **I. Velicogna**, M Rajeevan, 'Precipitation variability over India during the 20th and 21st centuries: investigating natural and anthropogenic drivers', *Climatic Change* 172 (3-4), 37.

PJ93- 2021 – Mishra S., Rupper S., Kapnick, Casey, Chan H., Ciraci E., Haritashya U., Hayse J., Kargel J., Kayastha R., Krakauer N., Kumar S., Lammers R., Maggioni V., Margulis S., Olson M., Osmanoglu B., Qian Y., Richey S., Rittger K., Rounce D., Shean D., **Velicogna I**., Veselka T., Arendt A. 'Grand Challenges of Hydrologic Modeling for Food-Energy-Water Nexus Security in High Mountain Asia', *Frontiers of Sciences* DOI: 10.3389/frwa.2021.728156

PJ92- 2021 – Du J, J.S. Kimball, J. Sheffield, **I. Velicogna,** M. Zhao, M. Pan, C. Fisher, H.E. Beck, J.D. Watts, G. A, E.F. Wood, 'Synergistic Satellite Assessment of Global Vegetation Health in Relation to ENSO-induced Drought and Pluvials' *J. Geophys. Res. Biogeosciences 126 (5), e2020JG006006* https://doi.org/10.1029/2020JG006006.

PJ91- 2021 – De Conto R., Pollard D., Alley R.B., **Velicogna I.,** Gasson E., N. Gomez, S. Rogstad, D. M. Gilford, E. L. Ashe, and R. E. Kopp 'The Paris Climate Agreement and future sea-level rise from Antarctica' *Nature* 593 (7857), 83-89.

PJ90- 2021 – Cheng D., W Hayes, E Larour, Y Mohajerani, M Wood, **I Velicogna**, Calving Front Machine (CALFIN): glacial termini dataset and automated deep learning extraction method for Greenland, 1972–2019, *The Cryosphere* 15(3), 1663-1675.

PJ89- 2021 – Y Mohajerani, S Jeong, B Scheuchl, **I Velicogna**, E Rignot, P Milillo, Automatic delineation of glacier grounding lines in differential interferometric synthetic-aperture radar data using deep learning, *Nature Scientific reports* 11 (1), 1-10

PJ88- 2021 – M Wood, E Rignot, I Fenty, L An, A Bjørk, M van den Broeke, C Cai, **I. Velicogna**, ... Ocean forcing drives glacier retreat in Greenland, Science Advances 7 (1), eaba7282

PJ87- 2021 – M Zhao, A Geruo, J Zhang, **I Velicogna**, C Liang, Z Li, Ecological restoration impact on total terrestrial water storage, *Nature Sustainability* 4 (1), 56-62

PJ86- 2021 – BD Hamlington, AS Gardner, E Ivins, JTM Lenaerts, JT Reager, **I. Velicogna**, ... Understanding of contemporary regional sea-level change and the implications for the future, *Reviews of Geophysics* 58 (3), e2019RG000672

PJ85- 2020 – Jakobsson, M. et al., The International Bathymetric Chart of the Arctic Ocean Version 4.0, *Nature Scientific Data*, 7, 176.

PJ84- 2020 – Sutterley T., **Velicogna I**., Hsu C., 'Self-Consistent Ice Mass Balance and Regional Sea Level from Time-Variable Gravity', *Earth and Space Science,* 7, e2019EA000860.

PJ83- 2020 – Ciraci' E., **Velicogna I.,** Swenson S., 'Continuity of the mass loss of the World's Glaciers and Ice Caps from the GRACE and GRACE Follow-On missions.', *Geophys. Res. Lett.* 47, e2019GL086926.

PJ82- 2020 – **Velicogna I.,** Mohajerani Y., A G., Landerer F., Mouginot J., Noel B., Rignot E., Sutterley T., van den Broeke M., van Wessem J., Wiese D.**,** 'Continuity of ice sheet mass loss in Greenland and Antarctica from the GRACE and GRACE Follow-On missions', *Geophys. Res. Lett*. 47, e2020GL087291.

PJ81- 2020 – Madani, N., Kimball J., Parazoo N., Ballantyne A., Tagesson T., Lucas J., Reichle R., Palmer P., **Velicogna, I.,** Bloom, A., Saatchi S., Liu Z., A G., 'Below-surface water mediates the response of African forests to reduced rainfall' *Environ, Res. Lett*., https://doi.org/10.1088/1748-9326/ab724a.

PJ80- 2020 – A G., **Velicogna I**., Kimball J., M. Zhao, Du J., Kim Y.; Colliander A., Njoku E., 'Satellite detection of varying seasonal water supply restrictions on grassland productivity in the Missouri basin, USA', *Rem. Sens. Environ.*, 239, 111623.

PJ79- 2019 – Shepherd A., Ivins E., Rignot E., Smith B., van den Broeke M., **Velicogna I. et al**.,, 'Mass balance of the Greenland Ice Sheet, 1992-2018', *Nature*, 579, 233–239.

PJ78- 2019 – Du J, Kimball JS, **Velicogna I,** Zhao M, LA Jones, JD Watts, Y Kim., 'Multi-component satellite assessment of drought severity in the Contiguous United States from 2002 to 2017 using AMSR-E and AMSR2', *Water Resources Research*, 55(7), 5394–5412.

PJ77– 2019 – Lu, A. et al., Bathymetry of southeast Greenland from oceans melting Greenland (OMG) data, *Geophys. Res. Lett.,* 46 (20), 11197–11205

PJ76- 2019 – Mohajerani Y., **Velicogna I,** Rignot E, 'Evaluation of Regional Climate Models using Regionally-Optimized GRACE Mascons in the Amery and Getz ice shelves basins, Antarctica', *Geophys. Res. Lett.,* 46(23), 13,883–13,891.

PJ75- 2019 – Sutterley T., **Velicogna I**., 'Improved Estimates of Geocenter Variability from Time-Variable Gravity and Ocean Model Outputs', *Remote Sensing*, 11 (18), 2108.

PJ74- 2019 – Tapley B., Watkins M., Flechtner F., Reigber C., Bettadpur S., Rodell M., Sasgen I., Famiglietti F., Landerer F., Chambers D., Reager J., Gardner A., Save H., Ivins E., Swenson S., Boening C., Dahle C., Wiese D., Dobslaw H., Tamisiea M., **Velicogna I.,** 'Contributions of GRACE to understanding climate change'. *Nature Clim. Change*, 9, 358–369.

PJ73– 2019 – Vignesh, PP, JH Jiang, P Kishore, H Su, T Smay, N Brighton, **I Velicogna**, 'Assessment of CMIP6 Cloud Fraction and Comparison With Satellite Observations'. *Earth and Space Science,* e465.

PJ72– 2019 – Basha G., MV Ratnam, KN Kumar, T Ouarda, P Kishore, **I Velicogna**, 'Long-term variation of dust episodes over the United Arab Emirates'. *J. Atmos. Solar-Terrestrial Phys.*187, 33-39.

PJ71– 2019 – Pangaluru K, **I Velicogna**, Y Mohajerani, E Ciracì, S Cpepa, G Basha, S. Rao, 'Soil Moisture Variability in India: Relationship of Land Surface–Atmosphere Fields Using Maximum Covariance Analysis', *Rem. Sens.*, 11 (3), 335.

PJ70– 2019 – Mohajerani Y., Wood M., **Velicogna I.,** Rignot E., 'Detection of Glacier Calving Margins with Convolutional Neural Networks: A Case Study'. *Rem. Sens.*, 11(1), 74; doi:10.3390/rs11010005.

PJ69– 2018 – Ciraci' E., **Velicogna I.**, Sutterley T., 'Mass Balance of Novaya Zemlya Archipelago, Russian High Arctic, Using Time-Variable Gravity from GRACE and Altimetry Data from ICESat and CryoSat-2'. *Rem. Sens.,* 10(11), 1817.

PJ68– 2018 – Cazenave et al., 'Global Sea Level Budget 1993–Present', *Earth Syst. Sci. Data, 10(3),* 1551-1590.

PJ67– 2018 – Sutterley T., **Velicogna I.,** Fettweis X., Rignot E., Noel B., van den Broeke M., 'Evaluating regional atmospheric climate models in the ablation zone of the Greenland Ice Sheet using satellite and airborne laser altimetry data', *Geophys. Res. Lett.,* 45(16), 8324-8333.

PJ66– 2018 – Abernethy, R., Ackerman, S.A., Adler, R., Albanil Encarnación, A., Aldeco, LS ... State of the climate in 2017, *Bull. Am. Meteorol. Soc.* 99 (8), Si-S310

PJ65– 2018 – Mohajerani Y, **Velicogna I.**, Rignot E, 'Mass loss of Totten and Moscow University glaciers, East Antarctica, using regionally-optimized GRACE mascons', *Geophys. Res. Lett.,* 45(14), 7010-7018.

PJ64– 2018 – Kishore P., **Velicogna I.,** Sutterley T.,Mohajerani Y., Ciraci' E., G. N. Madhavi,'A case study of mesospheric planetary waves observed over a three-radar network using Empirical Mode Decomposition', *Annales Geophysicae*, 36 (3), 925 -936

PJ63– 2018 – Basha G, Kishore P., Venkat Ratnam M., Ravindrababu S., **Velicogna I**., Jiang JH, Ao CO,' Global Climatology of Planetary Boundary Layer top obtained from multi satellite GPS RO observations', *Clim. Dyn.,* doi:10.1007/s00382-018-4269-1, 2018.

PJ62– 2018 – Shepherd A, Ivins E, Rignot E, Smith B, van den Broeke M, **Velicogna I.,** et al., 'Mass balance of the Antarctic ice sheet from 1992 to 2017', *Nature*, 558, 219-222, 2018.

PJ61- 2017 – M. Tedesco, J. E. Box, J. Cappelen, R. S. Fausto, X. Fettweis, K. Hansen , M. S. Khan , T. Mote, I. Sasgen, C. J. P. P. Smeets, D. van As, R. S. W. van de Wal***, I. Velicogna***, 'Greenland Ice Sheet" in State of the Climate in 2016*. Bull. Amer. Meteor. Soc*., 98 (8), S140-S153. DOI: 10.1175/2017BAMSStateoftheClimate.1, 2017.

PJ60- 2017 – Kishore P., Janapati J., Pay-Lian J., **Velicogna I.**, Sutterley S., Ciraci' E., Mohajerani Y. Balaji Kumar S., 'Investigation of Kelvin wave periods during Hai-Tang typhoon using Empirical Mode Decomposition'. *J. Atm. Solar Terrestrial Phys.* doi: 10.1016/j.jastp.2017.07.025.

PJ59- 2017 – Hsu C., **I. Velicogna**, "Detection of sea level fingerprints derived from GRACE gravity data", *Geophys. Res. Lett.,* 44, 8953–8961.

PJ58- 2017 – Bhanja S., A. Mukherjee, M. Rodell, Y. Wada, S. Chattopadhyay, **I. Velicogna**, K. Pangaluru, J. Famiglietti, " Groundwater rejuvenation in parts of India influenced by water-policy change", *Nature Sci. Rep.*, Scientific Reports (7), 7453.

PJ57-2017 – Basha S., Kishore P., Ratnam M., Jayaraman A., AghaKouchak A., Ouarda T., **Velicogna I.**, 'Historical and Projected Climate Change over India: Contribution of Natural and Anthropogenic Forcings". *Nature Sci. Rep.*, doi:10.1038/s41598-017-02130-3.

PJ56- 2017 – A G., **Velicogna I**., Kimball J., Du J., Kim Y.; Colliander A., Njoku E., 'Satellite-observed changes in vegetation sensitivities to surface soil moisture and total water storage variations since the 2011 Texas drought', *Environ. Res. Lett.*, doi: 10.1088/1748-9326/aa6965.

PJ55-2017 – Zhao M, A G., **Velicogna I**., Kimball J., 'A global gridded dataset of GRACE drought severity index for 2002-2014', *J. Hydrometeor.*, doi.org:10.1175/JHM-D-16-0182.1.

PJ54-2017 – Zhao M, A G., **Velicogna I**., Kimball J., 'Evaluation of a GRACE observation based terrestrial drought severity index in the Continental United States', *J. Clim*., doi:10.1175/JCLI-D-16-0458.

PJ53-2016 – Bhanja S., Mukherjee A., Saha D, **Velicogna I.,** Famiglietti J. 'Validation of GRACE based groundwater storage change using in-situ groundwater level measurements in India'. *J. Hydrology*, 543, Part B, 729–738.

PJ52-2016 – Khazendar A., E. Rignot, D. Schroeder1, Seroussi H., Schodlok M., Scheuchl B., Sutterley T. **Velicogna I.,** Rapid submarine ice melting in the grounding zones of ice shelves in West Antarctica', *Nature Comm.* 7 (2016): 13243.

PJ51-2016 – M. Tedesco, J. E. Box, J. Cappelen, X. Fettweis, K. Hansen, T. Mote, C. J. P. P. Smeets, D. van As, R. S. W. van de Wal, **I. Velicogna**, and J. Wahr, 'Greenland ice Sheet' [in "State of the Climate in 2015"]. *Bull. Amer. Meteor. Soc*., 97 (8), S140-S153.

PJ50-2016 – Kishore P., **I. Velicogna**, M. Venkat Ratnam, Ghouse Basha, S. P. Namboothiri, J. H. Jiang, T. C Sutterley, G. N. Madhavi, S. V. B. Rao, 'Sudden stratospheric warmings observed in the last decade by satellite measurements and reanalysis', *Rem. Sens. Environ.,* 184, 263-275.

PJ49-2016 – Rignot E, Fenty I, Xu Y, Cai C, **Velicogna I**, Cofaigh CÓ, Dowdeswell JA, Weinrebe W, Catania G, Duncan D "Bathymetry data reveal glaciers vulnerable to ice-ocean interaction in Uummannaq and Vaigat glacial fjords, west Greenland." *Geophys. Res. Lett* , 43.6 2667-2674.

PJ48-2016 – Kishore P., G Basha, MV Ratnam, **I Velicogna**, T Ouarda, DN Rao, 'Evaluating CMIP5 models using GPS radio occultation COSMIC temperature in UTLS region during 2006–2013: twenty-first century projection and trends', *Clim. Dyn.*, 1-18.

PJ47-2016 – Lim LK, SD. Schubert, SMJ Nowicki, JN Lee, AM Molod, RI Cullather, B Zhao, **I. Velicogna**, 'Atmospheric summer teleconnections and Greenland ice mass variations in the last four decades', *Environ. Res. Lett. 11 (2), 024002.*

PJ46-2016 – Hansen J., M Sato, P. Hearty, R. Ruedy, M. Kelley, V. Masson-Delmotte, G. Russell, G. Tselioudis, J. Cao, E. Rignot, **I. Velicogna**, E. Kandiano, K. von Schuckmann, P. Kharecha, Allegra N Legrande, M. Bauer, K. Lo, "Ice Melt, Sea Level Rise and Superstorms: Evidence from Paleoclimate Data, Climate Modeling, and Modern Observations that 2°C Global Warming is Highly Dangerous", *Atmos. Chem. Phys. Discuss.* 15 (14), 20059-20179.

PJ45-2016 – Kishore P., S Jyothi, G Basha, SVB Rao, M Rajeevan, **I. Velicogna**, 'Precipitation climatology over India: validation with observations and reanalysis datasets and spatial trends'. *Climate Dyn.*, 46, 541-556.

PJ44-2015 – G. A, **Velicogna I.**, Kimball J., Youngwook K., 'Impact of changes in GRACE derived terrestrial water storage on vegetation growth in Eurasia', *Environ. Res. Lett.* 10 124024 doi:10.1088/1748-9326/10/12/124024.

PJ43-2015 – Famiglietti J.S., A. Cazenave, A. Eicker, J.T. Reager, M. Rodell, **I. Velicogna**, 'Satellites Provide the 'Big Picture' for Global Hydrology' *Science* 349 (6249), 684.

PJ42-2015 – Mayewski P.A., T. Bracegirdle, I. Goodwin, D. Schneider, N.A.N. Bertler, S. Birkel, A. Carleton, M.H. England, J-H. Kang, A. Khan, J. Russell, J. Turner and **I. Velicogna,** 'Potential for Southern Hemisphere climate surprises'. *J. Quaternary Sci.* 30 (5), 391-395.

PJ41-2015 – Basha GN, P. Kishore, MV Ratnam, T Ouarda, **I. Velicogna**, T Sutterley, 'Vertical and latitudinal variation of the intertropical convergence zone derived using GPS radio occultation measurements'. *Rem. Sens. Environ.* 163, 262-269.

PJ40-2015 – Madhavi, G. N., Kishore, P., Rao, S. V. B., **Velicogna, I.,** Basha, G., 'Two-day wave observations over the middle and high latitudes in the NH and SH using COSMIC GPSRO measurements'. *Advances in Space Res.*, doi:10.1016/j.asr.2014.09.032

PJ39-2014 – Sutterley T.C., **Velicogna I.**, Flament T., Rignot E., van den Broeke M., Mouginot J., 'Mass loss of the Amundsen Sea Sector confirmed by four independent techniques'. *Geophys. Res. Lett.* 41, 8421-8428.

PJ38-2014 – **Velicogna I.,** T.C. Sutterley, M. van den Broeke, 'Spatially varying ice mass acceleration of the polar ice sheets from GRACE'. *Geophys. Res. Lett.*, 41 (22), 8130-8137

PJ37-2014 – Kim, Y., Kimball, J. S., Zhang, K., Didan, K., **Velicogna, I.,** & McDonald, K. C., 'Attribution of divergent northern vegetation growth responses to lengthening non-frozen seasons using satellite optical-NIR and microwave remote sensing. *Int. J. Rem. Sens.*, 35(10), 3700-3721.

PJ36-2014 – Kishore P., **I. Velicogna**, 'Long-term Trends Observed in the Middle Atmosphere Temperatures Using Ground Based LIDARs and Satellite Borne Measurements'. *Annales Geophysicae,* 32 (3), 301-317

PJ35-2014 – Lupascu M, J. M. Welker, U. Seibt, X. Xu, **I. Velicogna**, D. S. Lindsey, C. I. Czimczik 'Water availability regulates magnitude, seasonality and sources of ecosystem respiration from a polar semi-desert in NW Greenland'. *Biogeosci.,* 11 (16), 4289-4304.

PJ34-2014 – Sutterley T.C., **Velicogna I.,** Csatho B., Broeke M., Evaluating Greenland glacial isostatic adjustment corrections using GRACE, altimetry, and surface mass balance data'. *Environ Res. Lett.,* 9 (1), 014004.

PJ33-2013 – Abraham JP, M. Baringer, N.L.Bindoff, T.Boyer, L.J. Cheng, J.A. Church, J.L. Conroy, C.M. Domingues, J.T. Fasullo, J. Gilson, G. Goni, S.A. Good, J.M. Gorman, V. Gouretski, M. Ishii, G.C. Johnson, S. Kizu, J.M. Lyman, A.M. Macdonald, W.J. Minkowycz, S.E. Moffitt, M.D. Palmer, A.R. Piola, F. Reseghetti, K. Schuckmann, K.E. Trenberth, **I. Velicogna**, J. Willis, 'A review of global ocean temperature observations: implications for ocean heat content estimates and climate change'. *Rev. Geophys.*, doi: 10.1002/rog.20022.

PJ32-2013 – Ratnam M., P. Kishore, **I. Velicogna**, 'Global distribution of pauses observed with satellite measurements'. *J. Earth Sys. Sci.*, doi: 10.1007/s12040-013-0278-y.

PJ31-2013 – Madhavi G. N. , P. Kishore, S.V.B. Rao, **I. Velicogna**, V. Sivakumar, 'Climatology and comparison study of stratosphere and lower mesosphere temperatures using satellite and reanalysis data sets'. *Int. J. Current Res. and Rev.,* 5(5): 17-42.

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PJ29-2013 – Yang C., **Velicogna I.,** Famiglietti J., Randerson J.,' Satellite observations of terrestrial water storage provide early warning information about drought and fire season severity in the Amazon', *J. Geophys. Res.*, doi: 10.1002/jgrg.20046.

PJ28-2013 – Church J., White N.J., Konikow L.F., Domingues C.M., Cogley J.G., Rignot E., Gregory J., van den Broeke M., Monaghan A., **Velicogna I.**, Correction to "Revisiting the Earth's sea-level and energy budgets from 1961 to 2008" *Geophys. Res. Lett.*, 40 (15), 4066-4066. DOI:10.1002/grl.50752

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PJ26-2012 – Kishore P., **Velicogna, I**., Ratnam MV, Jiang JH, Madhavi GN, 'Planetary waves in the upper stratosphere and lower mesosphere during 2009 Arctic major stratospheric warming', *Annales Geophysicae,* DOI: 10.5194/angeo-30-1529-2012.

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PJ24-2012 – Sasgen I., van den Broeke M., Bamber J., Rignot E., Sandberg Sorensen L., Wouters B., Martinec Z., **Velicogna I.,** Simonse S., 'Timing and origin of recent regional ice mass loss in Greenland', *Earth Planet. Sci. Lett.* 333–334, p 293–303.

PJ23-2011 – Church J., White N.J., Konikow L.F., Domingues C.M., Cogley J.G., Rignot E., Gregory J., van den Broeke M., Monaghan A., **Velicogna I.**, 'Revisiting the Earths sea-level and energy budgets from 1961 to 2008', *Geophys. Res. Lett.*, Vol. 38, L18601, doi:10.1029/2011GL048794.

PJ22-2011 – Rignot E., **Velicogna I.**, van den Broeke, M., Monaghan A., Lenaerts J., 'Acceleration of the contribution of the Greenland and Antarctic ice sheets to sea level rise', *Geophys. Res. Lett.*, Vol. 38, L05503, doi:10.1029/2011GL046583.

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PJ18-2010 – Rignot E., Koppes M., **Velicogna I.**, 'Rapid submarine melting of the calving faces of West Greenland glaciers', *Nature Geosci.*, Vol. 3, 187-191, doi: 10.1038/NGEO765.

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PJ15-2009 – **Velicogna I.**,' Increasing rates of ice mass loss from the Greenland and Antarctic ice sheets revealed by GRACE', *Geophys. Res. Lett.*, Vol. 36, L19503, doi:10.1029/2009GL040222.

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PJ13-2006 – **Velicogna I.,** Wahr J., 'Acceleration of Greenland Ice Mass Loss in Spring 2004', *Nature*, Vol. 443, 329-331, doi:10.1038/nature05168.

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PJ9-2004 – **Velicogna I.**, Wahr J., Hanna E., Huybrechts P., 'Short term mass variability in Green- land, from GRACE', *Geophys. Res. Lett.*, Vol. 32, L05501, doi:10.1029/2004GL021948.

PJ8-2004 – Wahr J., Swenson S., Zlotnicki V., **Velicogna I.**, 'Time-Variable Gravity from GRACE: First Results', *Geophys. Res. Lett.*, Vol. 31, L1150, doi:10.1029/2004GL019779.

PJ7-2004 – Perez-Gussinye M., Lowry A.R., Watts A.B., **Velicogna I.**, 'On the recovery of realistic effective elastic thickness structures using spectral methods', *J. Geophys. Res.*, Vol. 109, B10409, doi:10.1029/2003JB002788.

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PJ4-2002 – **Velicogna I.,** Wahr J.,'A method for separating Antarctic postglacial rebound and ice mass balance using future ICESat Geoscience Laser Altimeter System, Gravity Recovery and Climate Experiment, and GPS satellite data', *J. Geophys. Res.*, Vol. 107, 2263, doi: 10.1029/2001JB000708.

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PJ2-1999 – Accaino F., Cazzaro R., Cernobori L., Costa G., Della Vedova B., Kravania S., Marson I., Nicolich R., Panza G.F., Suhadolc P., Sarao A., Pellis G., Romanelli F., Vaccari F., **Velicogna I.**, 'Structure and physical properties of the tectosphere in the metastable Europe', *Plinius*, 650-667.

PJ1-1998 – Marson, I., Nicolich R., Cernobori L., Stoka M., Liotta D., Palmieri F., **Velicogna I.**, 'CROP-03 profile: a geophysical analysis of data and results', *Mem. Soc. Geol. Ital. 52*, 123-137.

Book Chapters

BC4-2020 – Meredith, M., M. Sommerkorn, S. Cassotta, C. Derksen, A. Ekaykin, A. Hollowed, G. Ko nas, A. Mackintosh, J. Melbourne-Thomas, M.M.C. Muelbert, G. Ottersen, H. Pritchard, and E.A.G. Schuur, 2019: Polar Regions. In: IPCC Special Report on the Ocean and Cryosphere in a Changing Climate [H.-O. Pörtner, D.C. Roberts, V. Masson-Delmotte, P. Zhai, M. Tignor, E. Poloczanska, K. Mintenbeck, A. Alegría, M. Nicolai, A. Okem, J. Petzold, B. Rama, N.M. Weyer (eds.)].

BC3-2017 – Remy F., Memin A., **Velicogna I,** 'Applications of Satellite Altimetry to Study the Antarctic Ice Sheet' in Satellite Altimetry over Oceans and Land Surfaces, CRC Press ISBN 9781498743457.

BC2-2013 - Observations: Cryosphere Supplementary Material, DG. Vaughan, JC. Comiso, I Allison, J Carrasco, G Kaser, R Kwok, P Mote, T Murray, F Paul, J Ren [...], D Robinson, M Sharp, N Shiklomanov, S Smith, S Stammerjohn, **I. Velicogna**, P Wadhams, A Worby, L Zhao, IPCC AR5, Climate Change 2013: The Physical Science Basis, Edited by Stocker TF, Qin D., Plattner G-K, Tignor M., Allen SK, Buschung J., Nauels A., Xia Y., Bex V., Midgley PM, 09/2013: chapter Cryosphere Supplementary Material: pages 45M1-45M10; Cambridge University Press. IPCC AR5

BC1-2007 - Bentley, C., Thomas R., **Velicogna I.,** 'Ice sheets' in Global Outlook for Ice and Snow ISBN: 9789280727999 United Nations Environment Program Publication.

Peer-reviewed Technical report

2024. National Academies of Sciences, Engineering, and Medicine. 2024. *Thriving on Our Changing Planet: A Midterm Assessment of Progress Toward Implementation of the Decadal Survey*. Washington, DC: The National Academies Press. https://doi.org/10.17226/27743.

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2011. "Legacies and Lessons of International Polar Year 2007-2008*". National Research Council of the National Academy of Sciences*. DOI 10.17226/13321.

GRANTS AND CONTRACTS

Total over all time raised more than $21M in funded grants

Funded:

- *NASA ROSES-23 NNH23ZDA001N-GRACEFO* "Improving ice sheet mass balance estimates using Laser Ranging Interferometer (LRI) and Microwave Instrument (MWI) GRACE-FO and GRACE data". P.I.: Isabella Velicogna. Dates: 01/15/2024-01/14/2028. $946K.
- *NASA ROSES-22 NNH22ZDA001N-FINESST* "Determining the role of extreme precipitation events on surface mass balance in East Antarctica from ICESat/GRACE data, climate models, and automated weather stations". P.I.: Isabella Velicogna. Dates: 09/01/2023-08/31/2026. $150K.
- *UCOP California Climate Action Seed and Matching Grants* "A fusion outlook product for predicting climate-water variation toward efficient decision making". P.I.: Isabella Velicogna. Dates: 08/01/2023-07/31/2025. $2.0M.
- *NASA ROSES-20 NNH20ZDA001N-ICESAT2* "Mass balance of Antarctica combining ICESat-2, GRACE and other data in an iterative approach". P.I.: Isabella Velicogna. Dates: 05/20/2021-05/19/2024. $484K.
- *NASA ROSES-19 NNH19ZDA001N-SMAP* "Impact of changes in the water cycle on vegetation productivity from SMAP, GRACE and other data for model evaluation and improvement". P.I.: Isabella Velicogna. Dates: 11/01/2019-10/31/2022. $498K.
- *UCOP AFD program* "Advancing Faculty Diversity in Recruitment". Co-P.I.: Isabella Velicogna. Dates: 12/31/2019-06/30/2021. $500K.
- *NASA ROSES-17 NNH17ZDA001N-CRYO* "Evaluation of ice sheet surface mass balance models in the ablation zone using ICESat, Operation IceBridge and other data". P.I.: Isabella Velicogna. Dates: 07/01/2018-06/30/2021. $461K.
- *NASA ROSES-16 NNH16ZDA001N-SLCT "*Quantifying and reducing uncertainty in future global and local sea-level estimates: linking physics, observations, and risk analysis to inform climate adaptation" P.I.: Isabella Velicogna. Dates: 09/19/2017-09/18/2020, $1.8M.
- *NASA ROSES-15 NNH15ZDA001N-HMA,* "Contribution of High Mountain Asia glacier changes to river basin hydrology using GRACE gravity and other data". PI: Isabella Velicogna. Dates: 8/1/2016-7/31/2019, $706K.
- *NASA ROSES-15 NNH15ZDA001N-SMAP,* "Characterization of regional drought impacts using satellite observations from SMAP and GRACE". PI: Isabella Velicogna. Dates: 9/1/2016-8/31/2019. $460K.
- *NASA ROSES-15 GRACE and GRACE-FO SCIENCE TEAM,* "Ice sheet contributions to global sea level from GRACE and GRACE Follow-On". PI: Isabella Velicogna. Dates:12/1/2015-11/30/2019, $604K.
- *NASA ROSES-14 IceBridge Research*, 'Reducing uncertainties in Greenland surface mass balance using IceBridge altimetry, GRACE data and regional atmospheric climate model outputs' PI: Isabella Velicogna. Dates: 12/14-11/17. $310K.
- *NASA ROSES-13 Terrestrial Hydrology Program (THP),* 'Quantifying recent drought and terrestrial water storage changes on boreal/Arctic vegetation and productivity using integrated satellite observations from MODIS, AMSR-E and GRACE'. PI: Isabella Velicogna. Dates: 01/14-12/17. $378K.
- *NASA ROSES-13 Sea Level Rise (SLR),* 'Land Contributions to Regional and Global Mean Sea Level Rise'. PI: Jay Famiglietti, Co-PI: Isabella Velicogna. Dates: 10/13-9/19, $1.3M.
- *NASA ROSES-13 Sea Level Rise (SLR),* 'Mass balance and bed topography datasets of ice sheets for sea level studies'. PI: Eric Rignot (UCI), Co-PI: Isabella Velicogna. Dates: 10/13-9/16, $1.2M.
- *NASA-GNSS Remote Sensing Science Team ROSES 11*, 'Accurate meteorology over Antarctica based on GPS-RO profiles for GRACE mass balance determination'. PI: David Bromwich (OSU), Co-P.I.: Isabella Velicogna. Dates: 10/2013-09/2016, $607K.
- *Gordon & Betty Moore Foundation Science Program* 'Fire Season and Drought Forecasting System Development'. Dates: 2012-2018 PI: James Randerson UCI. Co-PI Isabella Velicogna. $1.9M.

- *Jet Propulsion Laboratory - GRACE Follow-On Science Advisory Team*. 'Improving the accuracy of GRACE long term and interannual estimates of regional ice sheet mass balance'. PI: Isabella Velicogna, Dates: 2012-2015. $210K.
- *Center for Space Research - University of Texas, Austin – GRACE Science Team Member.* 'Assessment and validation of GRACE data products' Dates: 6/2012-9/2013. PI: Isabella Velicogna, $112K.
- *Center for Space Research - University of Texas, Austin – GRACE Science Team Member.* 'Assessment and validation of GRACE data products' Dates: 09/2013-10/2014. PI: Isabella Velicogna, $114K.
- *NASA ROSES-07 Hydrology Program*. 'Terrestrial water storage changes and boreal drought under a warming climate: investigation with multi-sensor satellite observations, in-situ data, and hydro-ecological process modeling'. PI: Isabella Velicogna. Co-P.I.: Kyle McDonald (JPL) and John Kimball (U. Montana). Dates: 06/09-05/13, $600K.
- *NASA ROSES-09 Inter-Disciplinary Science (IDS)-Program*, 'Regional pattern of sea level change from the melting of the Greenland, Antarctic and Patagonia ice sheets derived from combined InSAR ice motion, reconstructed surface mass balance and GRACE time-variable gravity'. PI: Eric Rignot, Co-PI: Isabella Velicogna, John Wahr (U. Colorado) Dates: 08/10-07/13, $1.247M
- *NASA-NNH08ZDA001N Modeling, Analysis and Prediction (MAP) Program*, 'A three-dimensional, high-resolution flow model of the Greenland ice sheet: validation and prediction'. PI: Eric Rignot, Co-PI Isabella Velicogna, Eric Larour (JPL). Dates: 01/09-12/12, $962K.
- *NASA-Hydrology Program*, 'Mass changes in Earth's global water reservoirs'. PI: James Famiglietti, Co-PI: Don Chamber (U. Texas), Steve Nerem (U. Colorado), Isabella Velicogna. Dates: 2007-2012, $1,034K
- *NASA Hydrology Program*, 'The Eurasian fresh water cycle from GRACE and in-situ hydrologic networks: distinguishing water behavior in permafrost and non-permafrost regions'. PI: Isabella Velicogna, Co-PI: Larry Smith (UCLA). Dates: 2008-2012, $271K.
- *NASA ROSES-07 Cryosphere Program*. 'Interpretation and uncertainties in ice sheet mass balance and seasonal variability from GRACE data'. PI: Isabella Velicogna. Co-PI: Mike Watkins (JPL). Dates: 06/08-05/12, $457K.
- *NASA ROSES-06 Solid Earth Program, '*Geophysical applications of GRACE', PI: John Wahr (U. Colorado), Co-PI: Isabella Velicogna. Dates: 07/07-10/11, $767K.
- *NASA ROSES-05 Cryosphere Program*, 'Mass balance of the polar ice sheets'. PI: John Wahr (U. Colorado), Co-PI: Isabella Velicogna. Dates: 09/07-09/11, $304K.
- *NSF-OPP*, 'Collaborative Research : Calculation of Antarctic gravity field from GRACE satellite data and comparison with independent measurements'. PI: Isabella Velicogna, Co-PI: Garry Karner (LDEO), Robin Bell (LDEO). Dates: 2004-2008, $455K.
- *NASA SENH Research and Applications Program*, 'Geopotential heights for GRACE de-Aliasing: comparison of SAC-C, CHAMP and GRACE occultation data with global circulation models', PI: Isabella Velicogna Co-PI: George Hajj (JPL). Dates: 2004-2008, $245K.

PROFESSIONAL ACTIVITIES AND INTERNATIONAL COMMITTEE MEMBERSHIPS

- 2026-Present - National Academies of Sciences Decadal Survey for Earth Science and Applications from Space 2028-2037 (ESAS 2028): Panel on Polar Ice Sheets, Glaciers, and Sea Ice Dynamics ("Cryosphere"). Nominated/Appointed Committee Member
- 2025 - Committee on Earth Science and Applications from Space (CESAS) National Academy of Science, Meeting Participant Invited, October 27-29
- 2024-Present - Editorial Board - Advisory Committee- ARC Geophysical Research.

- 2023-2024 - National Academies of Sciences committee on 'Midterm Assessment of the 2018 Decadal Survey for Earth Science and Applications from Space' Nominated/Appointed.
- 2022-2025 - Chair Electorate Nominating Committee, Atmospheric and Hydrospheric Sciences, Annual Election of AAAS Officers. Elected.
- 2021-Present -NASA Ice, Cloud, and land Elevation Satellite 2 (ICESat-2) mission, Science Team Member. The mission explores for measuring ice sheet mass balance, cloud and aerosol heights, as well as land topography and vegetation characteristics.
- 2021 Spring - Consulted with the "House Science, Space, and Technology Committee, the Space and Aeronautics Subcommittee" on NASA's role in climate science, particularly the role of satellite data in measuring climate change and its impacts on Earth systems.
- 2021-Present -Member of the Executive Editorial Board, of Environmental Research: Climate (ERCL), IOP (Institute of Physics) Publishing.
- 2020-2025 -Electorate Nominating Committee, Atmospheric and Hydrospheric Sciences, Annual Election of AAAS Officers. Elected.
- 2019-2020 -National Academies of Sciences Workshop on "Groundwater Recharge and Flow: Approaches and Challenges for Monitoring and Modeling using Remotely Sensed Data". Invited Expert Speaker/Panelist. Nominated.
- 2020-Present -Associate Editor Elementa, University of California Press.
- 2019-2022 -Editor Geosciences, MDPI.
- 2018-2020 -National Academies of Sciences committee on "Evolving the Geodetic Infrastructure to Meet New Scientific Needs". Committee Member. Nominated.
- 2020-2023 - AGU's Whitten Medal Selection Committee. Chair. Appointed.
- 2018-2020 - AGU's Whitten Medal Selection Committee. Member. Appointed.
- 2017-2021 - EGU's Vening Meinesz Medal Selection Committee. Appointed.
- 2017-2020 - Intergovernmental Panel on Climate Change (IPCC) Special Report on the Ocean and Cryosphere in a Changing Climate (SROCC), *Contributing Author*, Chapter 2: High Mountain Areas. *Contributing Author*. Appointed.
- 2017-2020 - Intergovernmental Panel on Climate Change (IPCC) Special Report on the Ocean and Cryosphere in a Changing Climate (SROCC), Chapter 3: polar Regions. *Contributing Author.* Appointed.
- 2013-2016 - Intergovernmental Panel on Climate Change Fifth Assessment (IPCC). *Contributing Author*, Working Group I , Chapter 3, "Cryosphere". Appointed.
- 2016-Present - NASA Sea Level Change Team (SLCT), *Science Team Member Appointed.*
- 2016-Present - NASA High Mountain Asia science Team (HiMAT), *Science Team Member Appointed.*
- 2016-Present - NASA Soil Moisture Active Passive (SMAP) satellite mission, *Science Team Member.*
- 2016-2017 - White House Office of Science and Technology Policy (OSTP) national Earth Observations Assessment (EOA 2016). Appointed as a Climate subject matter expert for Land ice.
- 2015-2016 - NASA- Earth Surface and Interior (ESI) Focus Area, to define of NASA-ESI strategic plan updated for the next 15+ years. *Committee member, appointed*.
- 2015-Present - Executive committee NASA/European Space Agency (ESA): Ice sheet Mass Balance Inter-comparison Experiment-2 (IMBIE-2) initiative. *Executive committee member*, and *discipline leader for time variable gravity. Appointed*.
- 2015/07 - Panelist for UCI-Earth System Science for His Holiness The Dalai Lama 80th Birthday Celebration. "Panel on the Effects of Climate Change and Taking Action to Resolve this Global Issue", Irvine (CA).

- 2015-Present - NASA/German Aerospace Center (DLR) *Science Team Member* of Gravity Recovery and Climate Experiment (GRACE) and GRACE Follow-On (FO) Satellite missions. *Appointed.*
- 2012-Present - NASA/DLR *Science Definition Team Member* of GRACE Follow-On satellite mission. *Appointed.*
- 2002-2015 - NASA/DLR *Science Team Member* of the GRACE-1 satellite mission. *Appointed.*
- 2006-2013 - Editor Geophysical Journal International
- 2013 - National Academy of Sciences (NAS) committee on 'Opportunities to Use Remote Sensing in Understanding Permafrost and Ecosystems using remote sensing to advance our understanding of permafrost status and trends as well as the ecological impacts of permafrost change'. *Invited Expert Panel member*, Nov 2013.
- 2013 - Scientific Committee on Antarctic Research (SCAR) Antarctic Climate 2100 Scientific Research Program for "Quantification of Antarctic Climate Variability", Nov 2013. *Invited Expert Panel member and guest speaker*, Invitation only.
- 2013 - Keck Institute for Space Studies (KISS), Caltech-JPL, *Invitee to the* "The Sleeping Giant: Measuring Ocean Ice Interactions in Antarctica" international workshop to study the "tipping point" of global sea level rise and develop scientific requirements for an observing system to monitor the Southern Ocean. Invited Expert member, Sep 2013.
- 2012-Present - International Union of Geodesy and Geophysics (IUGG) and the International Association of Geodesy (IAG). *Invited Expert committee member* on 'Future gravity missions science requirements for cryospheric research'.
- 2010-2013 - Intergovernmental Panel on Climate Change Fifth Assessment (IPCC AR5). *Contributing Author and Expert Reviewer*, Working Group 1, Chapter 4. Appointed.
- 2012-2013 -National Climate Assessment (NCA) 2013. *Invited contributor*.
- 2011-Present - International Association of Geodesy (IAG). *Invited Membe*r Inter-commission Committee on Theory (ICCT) - Commission 2 joint study group 'Theory and Inversion in Gravity-Solid Earth Coupling' (2011-Present).
- 2011-2014 -NASA/European Space Agency (ESA): Ice sheet Mass Balance Inter-comparison Experiment (IMBIE). *Participant. Appointed*.
- 2011 - National Academy of Science, *Panel Member* Workshop to produce a report on Legacies and Lessons of International Polar Year 2007-2008. *Appointed. Invitation only*.
- 2004-2010 -American Geophysical Union (AGU), *Member, Hydrology Remote Sensing Committee*.
- 2006-2007 -American Geophysical Union (AGU), *Chair, Hydrology Remote Sensing Committee*.
- 2006-2008 -American Geophysical Union (AGU), *Chair, Geodesy Fall Meeting.*
- 2006-2008 -American Geophysical Union (AGU), *Member, Geodesy Executive Committee*.
- AGU *Session Organizer and Convener* (Fall AGU 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2014, 2015, 2016)
- European Geophysical Society *Session Organizer and Convener* (2002, 2003; EGU 2004, 2005, 2006, 2007)
- 2007- International Union of Geodesy and Geophysics (IUGG) *Session Organizer and Convener*
- 2004-Present- Northern Eurasia Earth Science Partnership Initiative (NEESPI) *Science Team Member.*
- 2004-2012- NASA Energy and Water Cycle Study (NEWS) *Science Team Member*.
- 2004-2012- NASA Land Cover Land Use Change *Science Team Member*.

• 2002- Present- Ad hoc Manuscript Reviewer for J. Geophys. Res., Geophys. Res. Lett., Geophys. J. Int., J. of Hydrology, Earth Planet. Sci. Lett., Earth, Planets and Space, J. of Geodesy, J. of Glaciology, J. of Climate, Science, Nature, Nature climate change, Nature geoscience, the Cryosphere.

• 2003- Present Proposal Reviewer (for National and International Agencies) NASA; NSF; National Oceanic and Atmospheric Administration (NOAA); European Space Agency (ESA); Netherlands Organization for Scientific Research (The Netherlands); The Secretariat of the Top-level Research Initiative (Norway); Australian Antarctic Science (AAS) Program.

• 2007-Present- Review Board for Funding Agencies (*Panel Member)*: NASA Cryosphere Program. NSF Office of Polar Program.

PROFESSIONAL MEMBERSHIPS

2018-Present	American Meteorological Society (AMS)
2016-Present	American Association for the Advancement of Science (AAAS)
1999-Present	American Geophysical Union (AGU).
2004-Present	International Glaciological Society (IGS).
1995-Present	European Geophysical Union (EGU)

RECENT OUTREACH ACTIVITY

- Organized a public event for Women in STEM: "Women in STEM: A Conversation with NASA Climate Scientist Claire Parkinson" Dr. Claire Parkinson Senior Scientist in the Climate Change Division of the NASA Goddard Space Flight Center. The remote event had 145 attendees from across the US, Canada, Italy, France
- Co-organized a conference within the UCI/Newkirk Center lecture series 'Toward a Sustainable 21st Century', John Whiteley lead organizer.
- Interviews on BBC news and BBC world about climate change related topics.
- Interviews in Press and Medias about ice sheet mass balance.
- Served in the Scientific Committee for the exhibit "Poles in Peril" of the Aquarium of the Pacific, Long Beach, CA. (2008-2011)
- Participated in a Documentary on the GRACE satellite mission for the American Museum of Natural History, New York, NY.
- Participated in the development of a podcast series on climate change for Climate Central, a non-profit production company.
- Served in the Climate Science Rapid Response Team for selected invited scientists to provide rapid information to media and government officials on climate change issues, Ray Weymann organizer.

UC IRVINE SERVICES

• 2022-Present: Academic Planning Taskforce Promoting Student Wellness Appointed by Provost Hal Stern to consider student socialization and mental health in the post-Covid era and to make recommendations for the campus to respond to the serious challenges students and faculty face. Bi-weekly meetings up to 2023; and monthly thereafter. Delivered annual reports to the Provost. Developed a Resource Guide that is published on the provost website. Supporting Student Mental Health: A Resource Guide For Faculty and Staff https://provost.uci.edu/faculty-resources/supporting-student-mental-health-a-resource-guide-for-faculty-and-staff/

•2020-2023: Academic Planning Office Steering Committee On Reaffirmation (SCOR) of accreditation process for the Western Association of Schools and Colleges and its Executive Committee (SCOREX). Appointed by Provost Hal Stern, Vice Provost Roxane Silver and Associated Vice Provost Paul Kang. Office of Academic Planning & Institutional Research. These committees oversee, coordinate, and ensure successful completion of all aspects of UCI's reaffirmation of accreditation process for the Western Association of Schools and Colleges. The

group's charge includes: Become familiar with the timeline, phases, and milestones of WSCUC's accreditation process; Offer feedback on project and communications plans, as well as draft documents; Contribute to the identification, development, and promotion of institutional themes that will be the heart of the reaffirmation effort and most likely will have transformational impact for our students; Serve on a theme-specific subcommittee related to your expertise and/or area of interest; Facilitate broad campus participation in the reaffirmation accreditation process; and Participate in campus preparations for the Off-site Review (spring 2022) and Accreditation Visit (fall 2022). Additionally, members of SCOREX are asked to assess the WSCUC accreditation goals and propose strategies for our institution's response; Review UCI's findings of alignment with WSCUC Standards and propose follow-up actions, as necessary; and Troubleshoot process challenges and provide recommendations for their resolution. UC Irvine submitted its institutional self-study report on December 2022. In February 2023, the campus hosted a site visit by a team of reviewers selected by WSCUC from peer institutions to meet with students, faculty, staff, and administrators. In June 2023, the WSCUC Commission reviewed the team report, met with campus leadership, and concluded its review. In July 2023, the Commission reaffirmed UC Irvine's accreditation for 10 years.

- 2018-2021: Associate Dean of Graduate studies, Equity and Inclusion. The School of Physical Sciences has 626 graduate students, 166 tenure track faculty, 210 non-senate-researchers, post-doc, specialists-, and 110 staff. This was the first time a school on campus had an Associate Dean for Equity and Inclusion. I developed an agenda for the position, a network of collaborators, a structure for the collaborations, and a set of guidelines. In addition, I helped implement these guidelines and develop metrics for reporting on these guidelines and documenting progress. As an illustration of the significance of this work, the position has now been split into two separate Associate Dean positions.
- 2020-2021: Co-Organizer and panelist of campus wide Town Hall discussions on the theme of Dismantling Systemic Racism (4 events)
- Spring 2021-Present: Member of the UCI SEA Change implementation Advisory group appointed by Vice Chancellor for Diversity Equity and Inclusion, Doug Haynes. The goal of the committee is to serve as a consultative body where members help the Office of Inclusive Excellence (OIE) to gauge the potential impacts and assess progress associated with UCI SEA Change Action Plan. Quarterly meetings. Other members of this high level committee include VP Academic Personnel, Diane O'Dowd, and Dean of the School of Physical Sciences, James Bullock.
- Organized Campus Town Hall "A conversation about Anti-Asian Discrimination: reflections on how it is impacting us and what we can do to stop – with Dr. Phong Luong, Graduate Division Counselor. Town Hall to provide a venue for our community (students, post-doc, researchers, faculty, and staff) to discuss recent events and the Asian and Asian-American discrimination at large, to share how these events are impacting our lives and to discuss what we can do at the department/school and campus level to raise awareness, foster cultural change, and create a community that is supportive for all. Attendance from the entire campus. Organizer and Moderator. Campus wide attendance.
- 2020-Present: Steering committee -Dismantling systemic racism campus initiative, (2020-Present)
- 2020-Present: Dismantling systemic racism initiative Committee member and Facilitator for the group on "Implicit Bias and Anti-Racist protocols, education, and policies" that is tasked with identify protocols/procedures to limit the impact of implicit bias on any of the campus activities and process. https://sites.uci.edu/endracism/group-6-implicit-bias-and-anti-racist-protocols-education-and-policies/.
- Academic Planning Office Workgroup on Understanding Faculty Teaching Workloads, Appointed, (2020-2021).
- Chancellor's Inclusive Excellence Award Review Committee, with the charge to review all of the campus nominations made by the deans on campus and recommend the faculty for receiving the award to Vice Chancellor Douglas Haynes. Appointed, 2019.

- Search Committee for the Dean of the School of Physical Sciences. Appointed by the Provost (2018-2019).
- UCI Delegation to the International STEMM Equality Congress, The Netherlands, Oct 2018
- Senate Ad Hoc Diversity Committee (2018) to develop a charge for a Senate council on equity and inclusion. The Senate Cabinet and the Committee on Rules and Jurisdiction (CRJ) approved the "proposal to establish bylaw 90. equity and inclusion, council" implementing the committee defined charges
- Search Committee for Chairs of Earth System Science at UCI (2016).
- Search Committee for Chairs of Physics and Astronomy (2016).
- Dean review committee, School of Physical Sciences (2015-2016)
- UCI ADVANCE Program for Equity and Diversity, Equity Advisor, Appointed by Doug Haynes Vice Provost for Academic Equity, Diversity & Inclusion Equity Advisor: Faculty Assistant to the Dean, participates in faculty recruiting by approving search strategies and raising awareness of Best Practices; organizes faculty development programs, as well as addresses individual issues raised by women and underrepresented minority faculty (2015-2018).
- Foster the participation of UCI into the Jet Propulsion Laboratory (JPL) SURP program. SURP is a high-level agreement between JPL and academic organizations that recognizes broad connections between the two institutions in both science and engineering, enables seed funding of research projects, exchange of graduate students and postdocs, and opens a pipeline for hiring into JPL's organization. Leadership at JPL feels that the SURP arrangement glues JPL to its partners, and appears to align well with UCI's vision for growth.
- Evaluation of Mid-Career Hire proposals (2015-2016) for the School of Physical Sciences.
- Evaluation of High Impact Hire Proposals (HIHP), Fall 2015, Spring 2016, Fall 2016.
- Equity Advisor for the School of Physical Sciences, UCI, (2015-2018).
- Member of Strategic Planning Working Group 1 "Global Leadership To Solve 21st Century Challenges Through Excellence In Research And Teaching, Supported By Strategic Expansion And Institutional Innovation" part of the UCI strategic planning effort "UCI: Bright Past, Brilliant Future" (2014-2015).
- Member, Award Committee for Earth System Science (2015-2016-2017-2018).
- Member, Graduate Student Admission Committee for Earth System Science (2008-2010-2011-2012-2013-2014-2015-2016-2017).
- Board on Undergraduate Scholarships, Honors, and Financial Aids for the Fall 2013.
- Member, Search Committee for Faculty Position in Hydrometerology, Department of Engineering (2009–2010).
- Member, Search Committee for Faculty Position in Physical Climate, ESS (2011-2012).
- Served on 25 PhD Thesis committees.
- Coordination of Departmental Seminar Series. Invited and hosted a total of 10 guest speakers from outside UCI (2008–Present).
- Promote UCI female career in Science, Engineering and Mathematics by mentoring UCI female students and facilitating interactions with JPL-NASA scientists.
- Participate in NSF Integrative Graduate Education and Research Traineeship (IGERT) Proposal initiative (2008).
- Lead the definition of a UCI/University of Bari (Italy) Memorandum of Understanding to develop scientific collaboration and exchange of undergraduate and graduate students between the two institutions (2012-2013)
- Participate in the "Sea Level Rise - Vulnerability and Uncertainty " initiative by UCI Environment Institute to collaborate with the Jet Propulsion Laboratory and promote collaborations between the Schools of Physical Science, Engineering, Law and Humanities at UCI.

• Implement new and innovative educational material in all taught undergraduate courses to form a basis for future internet-based classes, to enhance student learning and to increase undergraduate student enrollment. Take leadership in implementing these changes in response to teaching and enrollment goals of the Department of Earth System Science.

MENTORING

Associate Project Scientist Supervisor

• Dr. Geruo A, ESS, UCI (2018-Present).
• Dr. P. Kishore, ESS, UCI (2008-2020).

Assistant Project Scientist Supervisor

• Dr. Prakash Jha, UCANR, UC Merced (2023-Present)
• Dr. P. Kishore, ESS, UCI (2008-2018).
• Dr. Geruo A, ESS, UCI (2016-2018).

Specialist Supervised

• Dr. Khalil Virji ESS, UCI (2024-Present).
• Mr. Arnav Agrawal ESS, UCI (2026-Present).

Post-Doctoral Fellows Supervised

• Dr. Natalia Pinzon Department of Human Ecology, UC Davis (2024-Present).
• Dr. Omid Bagheri ESS, UCI (2024-Present).
• Dr. Yara Mohajerani, ESS, UCI (2019-2020).
• Dr. Enrico Ciraci', ESS, UCI (2018-2020).
• Dr. Chia-Wei Hsu, ESS, UCI (2017-2018).
• Dr. Tyler Sutterley, ESS, UCI (2016)
• Dr. Geruo A, ESS, UCI (2013-2016).
• Dr. Jinjun Tong, ESS, UCI (2009-2012).

PhD Students Supervised

• Rylie Foster, ESS, UCI (2026-Present, Chair).
• Gulnaz Shalgumbayeva, ESS, UCI (2023-Present, PhD student, Chair).
• Swetha Cherukuri, ESS, UCI (2020-Present, PhD student, Chair).
• Collin Richardson, ESS, UCI (2021-2025, PhD student, Chair).
• Chia-Chun Liang, ESS, UCI (2018-2024, PhD student, Chair).
• Jinhyuk Kim, ESS, UCI (2019-2021, PhD student, Chair).
• Meng Zhao, ESS, UCI (2013-2021, PhD student, Chair).
• Zhuoya He, ESS, UCI (2016-2021 PhD student, Chair).
• Yara Mohajerani, ESS, UCI (2014-2019, PhD student, Chair).
• Enrico Ciraci, ESS, UCI (2012- 2018, PhD student, Chair,).
• Chia-Wei Hsu, ESS, UCI (2011-2017 PhD student, Chair, Completed).
• Tyler Sutterley, ESS, UCI (2010-2016, PhD student, Chair, Completed).
• Aimee Gibbons - ESS, UCI, (2013-2016, PhD student, Co-Advisor, Completed).
• George Azzari, ESS, UCI (2009-2010, PhD student, Chair).
• Dr. Adrienne Smith-Block, Lamont-Doherty Earth Observatory (LDEO), Columbia University, (2006-2011, Co-Advisor, Completed).
• Dr. Louise Sandberg Sørensen, National Space Institute, Geodynamics, (Technical University of Denmark) DTU-Space, Denmark, (2006-2010, Co-Advisor, Completed).
• Dr. Nicola Creati, Univ. of Chieti, Italy, (1999-2003, Co-Advisor, Completed).

Master's Thesis Students Supervised

• Hortense Roussillon, École Normale Supérieure - Paris - ENS - (2024-2025, Co-Chair).

• Enrico Ciraci', Dept. of Physics, University of Bari, Italy, (2010-2012, Co-Chair).
• Erin Delman- ESS UCI, (2015-2017, Completed, Committee member,)
• Rachel Resolarczyk, ESS, UCI (2008-2009, Completed, Chair).

Master's Students Research Supervision

• Enrico Tranquillini, Engeneering Dept, Politecnico di Milano, Italy (2020, research project advisor)
• Chiara Esposito, Engeneering Dept, Politecnico di Milano, Italy (2020, research project advisor)
• Michele Barucca, Engeneering Dept, Politecnico di Milano, Italy (2020, research project advisor)
• Sara Tonazzi, Engeneering Dept, Politecnico di Milano, Italy (2020, research project advisor)
• Paolo Fornoni, Engeneering Dept, Politecnico di Milano, Italy (2020, research project advisor)

Undergraduate Students Supervised

• Conner Clark, Information and Computer Sciences, UCI, research (2017-2018).
• Ian Wahl, Information and Computer Sciences, UCI, research (2017-2018).
• Justin Dobrusky, Information and Computer Sciences, UCI, research (2017-2018).
• Josue Alfaro, Physics Dept., UCI, Honor Thesis advisor (2008-2009).
• Catherine Hatch, Utah State University, Honor Thesis advisor, 2007.

Served On PhD Dissertation Committees Of

• Raphaelle Charrassin - ESS UCI, (2025-Present)
• Chris Chalmers- ESS UCI, (2024-Present)
• Lilli Manzo - ESS UCI, (2023- 2026, Complete)
• Jaehun Kim - ESS UCI, (2021-2024, Complete)
• Hanning Chen - ESS UCI, (2021-2024, Complete)
• Cynthia Wong - Chemistry UCI, (2020-2023, Complete)
• Alissa Matus - Chemistry UCI, (2020-2023, Complete)
• Shivani Ehrenfeucht - ESS UCI, (2020-2023, Complete)
• Tyler Pelle - ESS UCI, (2015-2021, Complete)
• Emily Kane- ESS UCI, (2015-2020, Complete)
• Michael Wood- ESS UCI, (2015-2019, Complete)
• Cilan Cai - ESS, UCI, (2014-2018, Complete).
• Hongju Yu - ESS, UCI, (2014-2018, Complete).
• Lu An - ESS, UCI, (2013-2017, Complete).
• Romain Millan- ESS UCI, (2015-2018, Complete)
• Hrishikesh Chandanpurkar - ESS, UCI, (2012-2016, Complete).
• Jamiat Nanteza - ESS, UCI, (2012-2016, Complete).
• Lawrence Collin - ESS, UCI, (2009-2014, Complete).
• Tsou Chun – Dept. of Civil and Environmental Engineering, UCI, (2009-2015, Complete).
• Jingjing Li – Dept. of Civil and Environmental Engineering, UCI, (2009-2015, Complete).
• Rebeka Sultana- Dept. of Civil and Environmental Engineering, UCI, (2009-2015, Complete).
• Karli Anderson - ESS, UCI, (2009-2013, Complete).
• JT Reager - ESS, UCI, (2007-2012, Complete).
• Dr. Qing Xia - Dept. Civil and Environmental Engineering, UCI, (2007-2011, Complete)

TEACHING

For all Classes I annually include the following updates:

- Each year I develop additional active learning/Interactive lecture materials
- Each year I update the course material to including new data/information/research on the different class subjects (Climate Change/Cryosphere etc.) e.g. new research on the state of the planet, new analysis of observational results due to longer records and updated research analysis and findings.

-Include current events, and new articles relevant to the course as well as changes in policies related to the Earth and climate based on improved science results.

Ongoing: I have developed and included in all classes, evidence based teaching strategies that have been shown to improve the retention and success of students from under-represented groups, as well as awareness of importance of diversity in the classroom. I continuously update classwork and lecture style and activities based on student performance and feedback ensure that all student will reach the class learning goals.

These approaches include:

-Weekly formative assessments (online quizzes, discussion activities, in class small group activities, in class quizzes-HW practice/discussion)

-Active learning activities with opportunities for peer instruction e.g. -student are asked to discuss question with their neighbor student, - in class group activities for which I have developed a structure so that every group member participate to the activity.

-Introduction of extra credit metacognition exercises to improve study skills e.g. Examine with the students how their study methods impacted the exam/HW outcome, analysis of which questions they lost point on and of how they can improve in the future, what different approaches they can use for studying and to reason through the question

-Increasing engagement by using different media to introduce concepts: using video and news articles about research topics to showing the impact of the research on everyday life and government policies and decision making. This includes world wide examples from different cultural settings and related to environmental justice issues

-Use of TA to provide more peer-based instruction, and mentorship for students who struggle in classroom environments.

Courses developed and taught at UC Irvine:

EarthSS 15 Introduction to Global Climate Change, Spring 2010, Spring 2011, Spring 2012, Spring 2013, Spring 2014 UCI. General Education class that I developed and practiced.

EarthSS 236 Advanced Remote Sensing, Spring 2008-2009. Elective, upper-level graduate courses.

EarthSS 112 Global Climate Change, Winter 2017, Spring 2018, Fall 2018, Spring 2020-2021-2022-2023-2024-2025 UCI. Elective, upper-level undergraduate, graduate course. I re-developed this course lecture and practice.

EarthSS 116 Data Analysis, Fall 2008, Fall 2009, Fall 2010, Fall 2011, Fall 2012, Winter 2024 UCI. Required course for majors in Earth and Environmental Science.

EarthSS 174 Ice in the Climate System, Winter 2014, Winter 2015, Winter 2016 UCI. Elective, upper-level undergraduate, graduate course. I developed this course lecture and practice.

EarthSS 202 Climate Change. Spring 2015, Spring 2016 UCI. Graduate.

EarthSS 215 Cryosphere, Winter 2023-2026. UCI. Elective, upper-level undergraduate, graduate course. I developed this course lecture and practice.

EarthSS 274 Ice in the Climate System, Winter 2014, Winter 2015, Winter 2016 UCI. Graduate.

EarthSS 191 Research for Undergraduate, Fall 2008, Fall 2009, Fall 2010, Fall 2011, Fall 2012 UCI. undergraduate Research.

EarthSS 280C Cryosphere Topics, Spring 2013, Fall/Winter/Spring 2014-2026 UCI. Elective graduate course.

EarthSS 280C Gravity Topics, Spring 2013, Fall/Winter/Spring 2014-2026 UCI. Elective graduate course.

FIELD EXPERIENCE

• Measurements of current, salinity and temperature profile in tidewater glacier fjords, Greenland (2008, 2015, 2018).
• Precision gravity measurements (Italy, 1997-1998)
• GPS campaign survey (Wabash Valley, midcontinent U.S., 1998)

INVITED PRESENTATIONS AT EDUCATIONAL AND GOVERNMENTAL INSTITUTIONS AND PROFESSIONAL MEETINGS

2026- NOAA's Geophysical Fluid Dynamics Laboratory formal seminar series. January 2027.
2026- Invited Speaker, California Climate Action Initiative Climate Tools for the Resilient California Summit, October 2026.
2026- Invited Speaker, Geodesy and Geophysics for a sustainable planet, University of Trieste, "Satellite measurement of time variable gravity to study the Earth fluid envelop"; 3 September 2026.
2026- Invited Speaker, 2026 NSF NGF Community Science Conference "From Human Workflow to AI Agent: a case study for Antarctic ice mass balance estimates" May 26-Jun 4 2026.
2026- Antarctic Futures Symposium 2026. "Building a digital twin for East Antarctica". 23-24 April 2026.
2025- Invited Speaker. California Climate Action Initiative Climate Tools for the Resilient California. "A fusion outlook product for predicting climate-water variation toward efficient decision making". Feb 2025
2024- Invited Speaker, NASA Solid Earth Team Meeting – CORE 2.0, developing next ten-year vision for Earth System Interior (ESI), "Gravity, geoid, and sea level" Washington DC, September 2024.
2024- Invited Speaker, Spotlight on Science, "Tracking ice and water in a changing climate with AI" Newport Beach Public Library, March 20, 2024.
2024- Invited Speaker, CA Department of Water Resources, "Improving S2S forecast, based on domain and driven AI", Sacramento, CA, February 2024.
2024- Invited keynote Speaker "Implementing domain driven AI to improve agricultural water management", Annual CII Conference – Fluid Futures: Adapting to Extremes, Sacramento CA, February 2024.
2023- Invited Speaker ISSI workshop on Remote Sensing in Climatology – Essential Climate Variables and their Uncertainties, "Essential Climate Variable for Ice sheets, and glaciers and ice caps", Bern, Swiss, November 2023.
2023- Invited Speaker NCAR "Monitoring climate change from space", Boulder, CO, October 2023
2023- Invited Speaker, Silicon Valley Webinar Landscape titled "Advances in innovation in Agritech and FoodTech, "Improving the accuracy of forecast of weather extremes using observation and end-user need driven Artificial Intelligence models, September 2023
2022- Invited speaker, University of Udine, Udine, Italy. "Towards a learning algorithm for forecasting extreme climate events and their impact on agriculture and society." March 2022
2021- Invited Speaker by the Team leaders of the Glaciers & Sea Level Collaboration Team of the Interagency Arctic Research Policy Committee (IARPC). IARPC brings together leaders from 16 agencies, departments, and offices across the U.S. federal government, to enhance research in the Arctic to represent the Gravity Monitoring of the Arctic glaciers, "State of the Greenland and Arctic glaciers from GRACE/GRACE-FO satellite observations "Nov 2021
2021- Invited speaker, Department of Civil and Environmental Engineering, UCI, CA. "Monitoring Climate Change from Space". Oct 2021
2021- Invited speaker for the NASA Program at Cypress College, Cypress, CA, "Monitoring climate change from space using a web of satellite sensors". Sep 2021
2020– John Nye Lecture of the American Geophysical Union, "Weighing climate change in the cryosphere from space", Dec 2020. The John F. Nye Lecture is presented annually and recognizes

recent accomplishments and outstanding ability to present exciting scientific research and findings to AGU member scientists outside the Cryosphere section.

2020- Invited Speaker European Polar Science Week, ESA-NASA Ice sheet Mass Balance Inter-comparison Experiment (IMBIE) project. "Continuity of the GRACE/GRACE-FO observations of Ice Sheet loss: new results". Invited as lead of the Gravity studies section of the European-US project

2020- International Glaciological Society Seminar series. "Ice Sheets, Sea level, and Diversity, Equity and Inclusion in Science". Oct 2020

2020- Advancing Faculty Diversity UCOP meeting "Advancing Faculty Diversity: School of Physical Sciences at UC Irvine Team" Oct 2020.

2020- Euro Science Open Forum (ESOF) 2020 - "Using remote sensing observations and climate models to inform decision making", Trieste, Italy, Sep 2020.

2019 Advancing Faculty Diversity UCOP meeting "Advancing faculty diversity in the school of physical sciences at UC Irvine", UC Davis, CA, Nov 2019.

2019 National Academies of Sciences Workshop on "Groundwater Recharge and Flow: Approaches and Challenges for Monitoring and Modeling using Remotely Sensed Data". Invited Expert Speaker/Panelist. "Monitoring groundwater from Space".

2019 – Invited speaker, Department Polytechnic of Engineering and Architecture of the University of Udine, 'Remote sensing studies of land water and ice mass from space: results and implications', Udine, Italy, Jan 2019.

2018 – Invited Speaker, Memento Ecole Polytechnic Federale De Lausanne Seminar Series, 'Time-variable gravity studies of ice sheets and glacier mass balance and partitioning of the water cycle in high mountain environment', Lausanne, Switzerland, Dec 2018

2018 – Invited Speaker, GRACE Science Team Meeting, 'Weighting ice from space with GRACE and GRACE-FO', Vandenberg, CA, May 2018.

2017 - Invited speaker, Heiland Lecture, Distinguished Lecture at Colorado School of Mines "Weighting climate change from space" Colorado School of Mines, Golden, CO, Nov 2017.

2017 - Invited speaker, at the International Conference on "High Altitudes meet High Latitudes: Globalizing Polar Issues Crans-Montana, Swiss, Sep 2017.

2017 - Invited speaker, European Geophysical Union (EGU), Vening Meinesz Medal Lecture "Weighting climate change from space" Vienna, Austria, Apr 2017.

2017 - Invited speaker, 3rd Blue Planet Symposium, "The changing cryosphere and impact on regional sea level" NOAA, College Park, Maryland, USA Jun 2017.

2016 - Invited speaker, Dept. of Earth and Planetary Science, **"**Weighting climate change from space**"** University of Harvard, CA. Apr. 2016.

2016 - Invited speaker, Dept. of Earth and Planetary Science, "Weighting climate change from space" University of Berkeley, CA. March 2016.

2015 - Invited speaker, National Academy of Science and Kavli Foundation, Frontiers of Science Symposium "Observing Earth from Space", Irvine, CA, Nov 2015.

2015 - Invited speaker, Center for Coastal Physical Oceanography and the Mitigation and Adaptation Research Institute (MARI) at Old Dominion University, "Regional sea level variability from ice sheets, glaciers and land hydrology" Norfolk, VA. Feb 2015.

2015 - Invited speaker, Greenland Surface Mass Balance Meeting, NASA's Goddard Space Flight Center, "Using GRACE to improve surface mass balance estimates", Baltimore, MD, Jan 2015

2014 - Invited speaker, Fall AGU 2014, Session: "Understanding spatial and temporal variability of sea level rise'. San Francisco, CA, Dec 2014.

2014 - Invited speaker, Dept. of Earth Sciences, University of Oxford, "Time variable gravity studies of ice sheet mass balance and regional sea level. Oxford, UK, Nov 2014.

2013 - Invited speaker for the Scientific Committee on Antarctic Research (SCAR) Antarctic Climate 2100 Scientific Research Program for Quantification of Antarctic Climate Variability, "Mass balance of the Antarctica ice sheet". Castine, Maine. Sept. 2013.

2012 - Invited speaker, Ice Sheet System Model (ISSM) Workshop, "Greenland and Antarctica ice mass changes from GRACE". University of California Irvine, Irvine, CA. Dec. 2012.

2012 - Invited speaker and panelist at the Italian Consulate and Italian Cultural Institute of Los Angeles to promote research work of Italian scientists in the US. Los Angeles, CA. June 2012,

2012 - Invited speaker, **'**Geodynamics Seminar Series' Woods Hole Oceanographic Institution. Woods Hole, MA. May 2012.

2011 - Invited speaker, UCI Smart Energy, Sustainable Environment (SE2) Tutorial, "Climate Change and the Diminution of the Polar Ice Caps", Beckman Center of the National Academies, Irvine, CA. Nov 2011.

2011 - Invited speaker and panelist**, '**Glacier Melt, Early Snowmelt & Sea Level Rise: Toward a Sustainable 21st Century Series' organized by the Newkirk Center for Science and Society. Beckman Center of the National Academies, University of California, Irvine. Irvine CA.

2011 - Invited speaker at the JPL workshop on 'Understanding Sea Level Rise: The Roles of Coupled Systems'. Pasadena, CA, Aug 2011.

2010 - Keynote speaker at the IPCC AR5 WG1 Workshop on Sea Level Rise and Ice Sheet Instabilities. 'Satellite- based Observations of Sea Level Rise from Large Ice Sheets', Kuala Lumpur, Malaysia, Jun 2010.

2010 - Invited speaker, Soil and Water Sciences Seminar Dept. of Environmental Sciences, UC Riverside, CA.

2009 - Invited speaker, Nuuk Climate Days, International Glaciological Society (IGS) Conference, Jul 2009.

2009 - Invited speaker, Joint DynaQlim/GGOS Workshop, Espoo, Finland, June 2009.

2008 - Invited keynote speaker at the Japanese-American Kavli Frontiers of Science Symposium - National Academy of Sciences symposium, May 2008.

2008 - Seminar series, Dept. of Physics, University of Udine, Udine, Italy, Aug 2008.

2008 - Environmental program on Global Change and Oceans organized by the Newkirk Center.

2008 - Whole Earth Seminar Spring, Dept. of Earth & Planetary Sciences University of California Santa Cruz, California.

2007 - Institute of Geophysics and Planetary Physics (IGPP), UCLA, California.

2007 - National Institute of Geophysics and Volcanology, Bologna, Italy.

2007 - Heiland Lecture Series, Department of Geophysics, Colorado School of Mines.

2007 - Seminar series, Dept. of Geophysics, Colorado school of Mines.

2007 - Jet Propulsion Laboratory-NASA, Division Seminar.

2006 - Dept. of Geophysics, Stanford University;

2006 - Dept. of Earth Sciences, University of Southern California.

2006 - Earth and Environmental Science Dept., New Mexico Tech.

2005 - Dept. of Geography, UCLA;

2005 - Dept. of Physics, New Mexico State University.

2004 - Jet Propulsion Laboratory (JPL)-NASA.

2003 - International Center for Science and High Technology, United Nation Industrial Development Organization (UNIDO);

2003 - University of Munich, Austria;– National Center for Atmospheric Research, Boulder;

2003 - National Institute of Oceanography and Experimental Geophysics, Trieste, Italy;

2003 - Program in Atmospheric and Oceanic Sciences (PAOS), University of Colorado.

2002 - Institute de Physique du Globe de Paris, (IPGP), France;

2002 - Dept. of Physics University of Toronto, Canada;

2002 - Institute of Geophysics, Austin TX;
2002 - Dept. of Civil and Environmental Engineering, Princeton University;
2002 - Dept. of Geomatics University of Calgary, Canada;
2002 - Jet Propulsion Laboratory (JPL)-NASA.
2001 - University of Udine, Italy.
1998 - Lamont-Doherty Earth Observatory, Columbia University;
1998 - Dept. of Geophysics, Massachusetts Institute of Technology (MIT);
1998 - Dept. of Geological Science, University of Utah;
1998 - Dept. of Geological Science, Indiana University.